

Surfare:
Your Digital Safety Assistant

Field	Community Service and Digital Inclusion
Website Proposal	https://surfare-nu.vercel.app/
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Project Overview

Problem Statement	Artificial intelligence is now within reach of almost anyone, and as these tools improve, the line between real and artificially generated content keeps getting thinner. AI-created or AI-enhanced messages, images, and links can look polished and trustworthy, and many people mistake them for the real thing. Globally, over half of adults experienced a scam in the past year, and nearly a quarter lost money (Global Anti-Scam Alliance [GASA] & Feedzai, 2025). In the Philippines, 84% of surveyed adults encountered a scam attempt, mostly through messaging apps and social media (GASA, 2026; Philippine News Agency [PNA], 2026). Confidence offers little protection: although most adults feel able to recognize a scam, many still fall for one, and the large majority of victims report harm to their mental well-being (PNA, 2026). Existing protection tools often require a stable internet connection, rely on technical language, or focus only on links while overlooking messages and images, leaving many users without understandable guidance at the moment they need it.



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	Therefore to conclude the need for a digital safety tool that can assess suspicious messages, links, and images, work at a basic level offline, use online threat checking when available, and explain its findings in simple language. Surfare addresses this need as a hybrid, AI-powered digital safety assistant.
Target Users	**Primary:** - Adults who need accessible guidance in recognizing potential scams and digital deception. **Secondary** - Junior high school, senior high school, and college students who encounter phishing, online fraud, and AI-generated content. - Parents, guardians, teachers, and community digital-literacy advocates.
Proposed Solution	**Surfare** is an offline-first, online-enhanced web application that helps users check suspicious messages, links, images, and QR codes. Basic checks run on the device using local detection rules, so they work without an internet connection. When online, threat-checking services add further verification. Surfare gives a simple risk level, explains the detected warning signs in plain language, and recommends safe next steps. Its camera coach applies principles of optics and illumination to flag blur, low light, and glare, helping users capture readable photos and scan QR codes safely. A friendly mascot guide, **Surfy**, answers follow-up questions through template-based responses tied to the detected signals, so users learn to recognize similar threats on their own. Future versions will add video analysis and upgrade Surfy into an AI-powered assistant.
General Objective	The proposed project aims to develop an offline-first, online-enhanced web application designed to assess



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	suspicious messages, links, images, or videos. The application aims to help users to recognize the online threats through on-device analysis, online threat-checking services, understandable explanations of warning signs, and gives clear explanations of why the content is flagged as suspicious. Additionally, it also provides safety guidance and recommendations to help the users to recognize similar threats in the future.
Specific Objectives	1. Develop an offline-first, online-enhanced web application that allows users to assess suspicious messages, links, images, or videos using locally stored detection rules even without internet connection. 2. Integrate online threat-checking services that provide additional information about suspicious contents when the internet is available. 3. Assess suspicious messages to identify phishing, scams, misleading statements, impersonation, and requests for personal information. 4. Assess suspicious links by determining the URLs for malicious patterns, suspicious redirects, and unsafe websites. 5. Develop an image-analysis module that extracts text from supported images using Optical Character Recognition (OCR), detects embedded QR codes, and analyzes the extracted text and decoded URLs using predefined rules to identify potential scam indicators and suspicious links. 6. Identify the file type to determine the appropriate analysis method and to gather relevant information from the submitted content. 7. Identify and explain warning signs by presenting the indicators found during analysis and explaining why they may suggest online threats. 8. Develop a predefined question and answers that match the users questions with appropriate response.



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	9. Provide context-based safety recommendations that suggest actions, such as avoiding suspicious links, verifying information, and protecting personal information. 10. Present results in simple and understandable language to help the users with lack of cybersecurity knowledge to recognize the threats. 11. Protects the users privacy by limiting the collection and storage of submitted content and informing users when the information may be transmitted to external threat checking. 12. Communicate assessment limitations by informing the users when the evidence is insufficient to determine whether the content is safe, suspicious, or malicious. 13. Check the application's performance and functionality by assessing the online and offline features, response time, file handling, results, and the ability to provide explanations through the different connectivity conditions. 14. Assess the usefulness and understandability of the applications explanation and guidance in helping the users in identifying the threat.

Project Features

Key Features	1. **Multi-Content Threat Assessment** - Allows users to submit suspicious text messages, URLs, images, and supported QR codes for analysis. Each content type is processed using its appropriate detection method. 2. **Offline-First Detection** - Uses locally available detection rules to identify predefined warning signs without an internet connection. Displays which checks are available offline and which require connectivity. 3. **Online Threat Verification**



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- Uses an external threat-intelligence service to check supported URLs against known malicious or phishing indicators when internet access is available.

4. **Risk Classification and Results**
   - Presents a clear risk category, such as Low Risk, Suspicious, or High Risk, together with the detected indicators and a note when the evidence is insufficient.
5. **Explainable Threat Indicators**
   - Explains why content was flagged, such as urgent payment demands, requests for passwords or OTPs, suspicious URL structures, or mismatched sender identities.
6. **Surfy: Guided Safety Assistant**
   - Lets users ask predefined follow-up questions about the assessment. Surfy provides responses linked to detected warning signs and explains recommended precautions.
7. **Physics-Informed Camera Coach**
   - Checks image quality for factors such as blur, insufficient illumination, and glare. Gives capture suggestions based on optics, light, and image clarity before a user submits an image or scans a QR code.
8. **Privacy and Secure File Handling**
   - Limits unnecessary collection and retention of submitted content, validates supported file types and sizes, and informs users when data may be sent to external services.
9. **Performance and Usability Evaluation**
   - Tests offline and online functionality, response time, supported file handling, detection behavior, and how well users understand the explanations and safety guidance.



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Proposed Technologies or Tools, if applicable	The proposed technologies and tools for Surfare are grouped according to their system functions: 1. **Frontend Development** – React.js, Vite, JavaScript, and Tailwind CSS for building a responsive and accessible user interface. 2. **Offline Functionality** – Progressive Web App (PWA), vite-plugin-pwa, and Workbox Service Workers for caching the application, detection rules, and required OCR resources for offline use after initial download. 3. **Message Threat Detection** – JavaScript-based contextual detection rules covering English, Filipino, and Taglish to identify combinations of scam indicators, including urgency, impersonation, advance-payment demands, and requests for passwords or OTPs. 4. **URL Analysis and Online Verification** – The JavaScript URL API and *tldts* for examining actual domains and suspicious URL structures, combined with Google Safe Browsing API for checking known unsafe links when online, subject to its free noncommercial usage terms and quotas. 5. **Image and QR Code Analysis** – Tesseract.js for extracting text from images and jsQR for decoding QR codes. Extracted text and URLs are assessed using the same message and URL detection modules. 6. **Camera and Image Quality Analysis** – Web Camera API, Canvas API, and OpenCV.js for assessing blur, brightness, and potential glare, and applying image preprocessing to support text extraction. 7. **Backend and API Integration** – Cloudflare Workers, within its free-tier limits, for securely handling online threat-checking requests and keeping API credentials outside frontend code.



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	8. **Risk Classification and Explanations** – A JavaScript-based decision engine that combines detected indicators and available threat-intelligence results to present risk categories, plain-language explanations, and an insufficient-evidence result when necessary. 9. **Surfy: Guided Safety Assistant** – Template-based JavaScript responses linked to detected warning signs and predefined follow-up questions to provide relevant safety guidance. 10. **Detection Evaluation** – A separate labeled test set of legitimate and scam content in English, Filipino, and Taglish to measure precision, recall, false positives, OCR performance, and online and offline functionality.
References	**Global Anti-Scam Alliance & Feedzai**. (2025, October 7). Global State of Scams Report 2025. https://www.gasa.org/post/global-scams-on-the-rise-over-half-of-adults-worldwide-report-scam-encounters **Global Anti-Scam Alliance. (2026)**. State of Scams in the Philippines 2026. https://www.mastercard.com/global/en/news-and-trends/Insights/2026/gasa-state-of-scams-in-the-philippines-2026.html **Philippine News Agency. (2026, October 12).** 84% of Filipino adults encountered scam attempts in past year. https://www.pna.gov.ph/article/84-of-filipino-adults-encountered-scam-attempts-in-past-year